Earthquake signals and Kilauea eruptions

UE24CS352A Machine Learning | Mini-project write-up

Prithvish Adhikari (PES1UG24AM206) · Piyush Shiv (PES1UG24AM191)

Problem statement

Can the location, depth, magnitude, and recent rate of earthquakes in Kilauea's East Rift Zone help identify an ongoing Pu'u O'o eruptive episode or estimate the time to the next eruption? We adapt these two tasks from the Stanford CS229 study by Mullet, Singh, and Wong (PEEVED) and add a time-ordered holdout that scores later earthquakes only.

Dataset and labels

The bundled catalogs contain 6,485 earthquake records and 49 eruption entries. Filtering to the East Rift Zone leaves 3,764 earthquakes; after a 30-day feature warm-up, the episode-48 coverage cutoff, and removal of records without a known next onset, 2,960 examples remain (304 eruptive, 10.3%). Features are longitude, latitude, depth, magnitude, and counts in the preceding 1, 7, and 30 days. Erupting labels use each catalog duration plus a 24-hour uncertainty buffer; time-to-eruption is measured in days. On the modeled rows these labels match the original PEEVED rule exactly. Applied to all 3,764 East Rift earthquakes that rule marks 22.4% as erupting (the paper reports about 20%); the lower 10.3% here comes from excluding 282 post-July-1986 events labelled only through episode 48's corrupted duration and the 522 warm-up earthquakes of January 1983.

Approach and implementation

We compare prior/mean baselines, logistic regression, random forests, Extra Trees, gradient boosting, class-conditional K-means, and a small multilayer perceptron. The original seven features remain available; classification also tests four motivated, strictly past-only temporal features. Scaling is fit on training data. Hyperparameters, feature set, and classification threshold are selected on development data, then evaluated once on the held-out test period. The chronological split uses 70% train, 20% development, and 10% test by time, a 30-day gap, and label-availability constraints. A random split is retained as a comparison. Code in src/data.py, src/models.py, and src/experiment.py writes the audit, metrics, predictions, and model bundles; demo.py predicts on a held-out earthquake.

Data quality decision

Episode 48 lists an implausible duration of 173,566,800 hours. We retain its onset as an exclusive cutoff and do not use the corrupted duration. The source also lists eruptions 35.5 and 35 out of order; preprocessing sorts onset-duration pairs together.

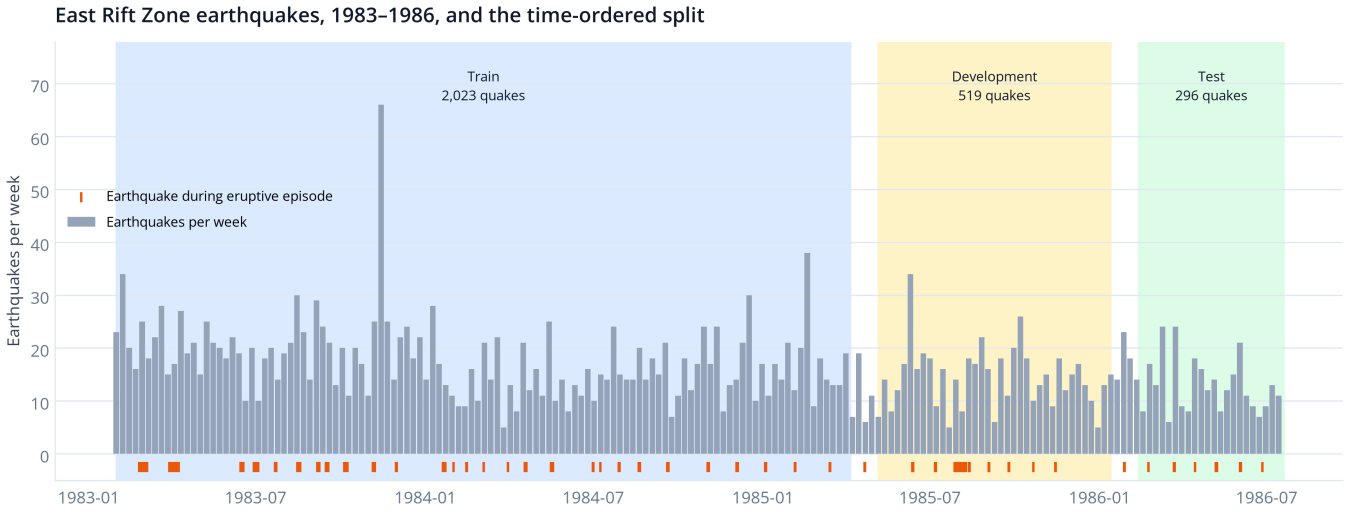


Figure 1. Weekly East Rift Zone earthquakes with eruptive-episode labels (orange ticks) and the time-ordered train / development / test windows.

Results and conclusion

The chronological test period is the primary measure of future-period performance.

0.484	-0.006	7.75 d	-0.179
classification AUROC	Cohen's kappa	regression RMSE	regression R-squared

Held-out classification

Development selected logistic regression with the expanded 11 feature set (C=10) and a development-F1 threshold of 0.611. On the later test window, AUROC was 0.484, average precision 0.206, and kappa -0.006. It identified 4 of 19 eruptive examples (recall 0.211) while marking 62 repose examples as eruptive (precision 0.061). At the fixed 0.5 threshold it would identify 11 eruptive examples with F1 0.101. The ranking score is near chance and the development-selected operating point produces many false alarms. The retained AUROC baseline would instead select gradient boosting (n=100,depth=2, development AUROC 0.603); the two procedures optimize different objectives, so the F1-selected procedure is not automatically an improvement.

Recall-first constraints

Under a development FPR cap of 0.10, recall-first selected a small neural network (MLP) at threshold 0.458; its development recall/FPR were 0.152/0.042, and its test recall/FPR were 0.053/0.090. Under a 0.20 cap it selected logistic regression at threshold 0.611; test recall/FPR were 0.211/0.224. The test FPR is not constrained because it was not used for selection, and neither strategy recovered the historical K-means positive count.

Controlled K-means threshold experiment

To isolate the historical model from model selection, we fixed class-conditional K-means (k=4) and the original seven features. The fixed 0.5 operating point found 12 of 19 test positives (recall 0.632) with 194 false positives. Development-F1 selection raised the threshold to 0.538 and found 9 positives. The development-FPR-constrained operating points found 1 and 6 positives for caps 0.10 and 0.20, respectively. These thresholds were selected without test labels; the K-means test AUROC and average precision are 0.524 and 0.096 for every row because thresholding changes decisions, not ranking scores. The fixed historical threshold and optimized thresholds are different operating procedures, and the results do not establish an improvement over the historical result.

Controlled K-means model improvement

We then evaluated k in 2, 3, 4, 5, 6 and 8 with raw and train-standardized original-seven-feature inputs and two positively oriented distance scores. Development AUROC selected k=3,scaling=raw,score=relative_distance (development AUROC 0.668, average precision 0.221). At its development-F1 threshold, the frozen test AUROC/AP were 0.515/0.068, with recall 0.158 and FPR 0.177. This is a development-selected ranking configuration, not evidence of reliable future-period forecasting; the test ranking can differ substantially from development performance.

Held-out time-to-eruption regression

Development selected the mean baseline. It achieved RMSE 7.75 days, MAE 6.60 days, and R-squared -0.179. All tested regression families had negative test R-squared, so none explained more variation than a test-period mean reference. These data do not support useful time-to-eruption forecasts.

What the random split changes

With a stratified random 70/20/10 split (as in the paper), development selected a random forest for classification (test AUROC 0.685, kappa 0.185) and a random forest for regression (RMSE 11.01 days, R-squared 0.160). The paper's random forest reported test kappa 0.52 and AUROC 0.71 (its text quotes kappa 0.42) and R-squared 0.38 on a standardized target. Neighboring earthquakes can appear in both training and test sets under a random split, so it is not an estimate of performance on later periods.

Conclusion

With seven catalog features and a small, imbalanced time series, these models do not forecast eruption timing reliably. The classification score has limited ranking power and low precision, while regression does not beat a mean reference by R-squared. Future work should test longer catalogs and independent signals such as ground deformation, gas emissions, and focal mechanisms. This is a classroom analysis, not an operational hazard tool.

Sources: Mullet, Singh, and Wong, “Predicting Eruptive Events at Volcanoes from Earthquake Data,” Stanford CS229, Fall 2019; original code and catalogs: github.com/bmullet/PEEVED (earthquakes: ANSS catalog via WOVOdat; eruptions: Hawaii Center for Volcanology). Results are this project's own runs on those bundled files (scikit-learn 1.7.2).